

Module 4

1. What is the significance of MAC layer and in which position it is placed in the OSI model?

The MAC (Medium Access Control) layer is part of the Data Link Layer (Layer 2) in the OSI model.

Significance of MAC layer:

- Controls how devices access the wireless medium (who transmits and when)
- Defines the format of Wi-Fi frames (MAC frame structure)
- Handles addressing using MAC addresses
- Ensures reliable data transfer using acknowledgements (ACK)
- Manages connection processes like scanning, authentication, and association
- Supports security through key management and encryption
- Provides QoS by prioritizing traffic (voice, video, etc.)
- Helps in power saving for wireless devices
- Supports efficient data transfer using frame aggregation

2. Describe the frame format of the 802.11 MAC header and explain the purpose of each field

The 802.11 MAC frame consists of MAC Header + Frame Body + FCS.
The MAC header contains important control and addressing information.

Main fields in MAC Header:

- **Frame Control (2 bytes):**
Defines the type of frame (management/control/data) and flags like retry, power, etc.
- **Duration/ID (2 bytes):**
Specifies how long the channel will be reserved (used for avoiding collisions)
- **Address 1 (6 bytes):**
Receiver address (destination device)
- **Address 2 (6 bytes):**
Transmitter address (sender device)
- **Address 3 (6 bytes):**
Used for source/destination/BSSID depending on frame type
- **Sequence Control (2 bytes):**
Helps in ordering frames and detecting duplicates
- **QoS Control (2 bytes) (optional):**
Used for traffic prioritization (QoS)
- **HT Control (4 bytes) (optional):**
Used in high throughput (802.11n and above)

Frame Body:

Contains actual data or management information

FCS (4 bytes):

Error detection using CRC

3. Please list all the MAC layer functionalities in all Management, Control and Data plane**Management Plane functionalities:**

- Scanning (Active and Passive)
- Client Association and Disassociation
- Authentication and Security Management
- QoS Management (traffic prioritization)
- Power Management (sleep/wakeup handling)
- Load Balancing (client distribution, band steering)

Control Plane functionalities:

- Medium Access Control (channel access using DCF, PCF, EDCA)
- Flow Control (rate control using acknowledgements)
- Collision Avoidance (CSMA/CA mechanism)
- Acknowledgement handling (ACK, Block ACK)

Data Plane functionalities:

- Data frame transmission between devices
- Frame Aggregation (A-MSDU, A-MPDU)

4. Explain the scanning process and its types in detail

Scanning is the process by which a Wi-Fi client (laptop/mobile) discovers available Access Points (APs) and their capabilities before connecting.

Types of Scanning:**1. Passive Scanning**

- The Access Point periodically sends Beacon frames (every ~100 ms)
- These beacon frames contain:
 - SSID (network name)
 - Supported data rates
 - Security information
 - Channel details
- The client listens on different channels and collects this information

Advantages:

- Less power consumption
- No extra traffic generated

Disadvantages:

- Slower (client has to wait for beacons)

2. Active Scanning

- The client actively sends a Probe Request frame
- Access Points reply with a Probe Response
- Probe request may be:
 - Specific (for a particular SSID)
 - Broadcast (for all networks)
- Client compares responses and selects the best AP

Advantages:

- Faster discovery
- More control over AP selection

Disadvantages:

- More power consumption
- Extra network traffic

5. Brief about the client association process

- **Scanning:**
The client searches for available Wi-Fi networks using passive (beacon) or active (probe) scanning
- **Authentication:**
The client sends an authentication request to the AP
The AP responds and allows the client (basic approval step)
- **Association:**
The client sends an association request
The AP replies with an association response
The client is now officially connected to the network
- **Data Communication:**
After association, normal data transfer begins

6. Explain each step involved in EAPOL 4-way handshake and the purpose of each keys derived from the process

The EAPOL 4-way handshake is used in WPA2/WPA3 to create secure encryption keys between the client and the Access Point (AP).

Steps in 4-way handshake:

- **Step 1:**
AP sends a random value (ANonce) to the client
Used to start key generation
- **Step 2:**
Client generates PTK (Pairwise Temporal Key) using:
 - ANonce
 - its own random value (SNonce)
 - MAC addresses
 Client sends SNonce to AP
- **Step 3:**
AP also generates the same PTK
AP sends GTK (Group Temporal Key) encrypted using PTK
- **Step 4:**
Client confirms successful key installation

Keys involved:

- **PTK (Pairwise Transient Key):**
Used for secure unicast communication (one-to-one data)
- **GTK (Group Temporal Key):**
Used for broadcast/multicast communication

7. Describe the power saving scheme in MAC layer and explore on the types of Power saving mechanisms

Wi-Fi devices (phones/laptops) save battery by switching between active and sleep modes. The Access Point (AP) temporarily stores data for sleeping devices and delivers it when they wake up.

How it works:

- Device informs AP that it is going to sleep mode
- AP buffers (stores) incoming data for that device
- AP sends Beacon frames containing TIM (Traffic Indication Map)
- Device wakes up periodically and checks if data is pending
- If data is available, device sends PS-Poll request
- AP delivers the stored data

Types of Power Saving mechanisms:

- **Legacy Power Save (PS mode):**
Device sleeps and wakes up periodically to receive buffered data using PS-Poll
- **U-APSD (Unscheduled Automatic Power Save Delivery):**
Device triggers data delivery by sending a frame
Used for real-time traffic like voice/video (better efficiency)
- **WMM Power Save (based on QoS):**
Advanced version of U-APSD with traffic prioritization
Improves battery life and performance

8. Describe the Medium Access Control methodologies

Medium Access Control (MAC) methodologies define how devices share the wireless channel and avoid collisions.

Main MAC methodologies:

1. DCF (Distributed Coordination Function)

- Uses CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)
- Steps:
 - Device checks if channel is free
 - If busy : wait
 - If free : wait random backoff time
 - When timer reaches 0 : transmit
 - Wait for ACK
- If no ACK : assume collision : retransmit

Key point:

Used in most Wi-Fi networks

2. PCF (Point Coordination Function)

- Controlled by Access Point
- AP gives permission to devices to transmit
- No collisions

Limitations:

- Complex and not widely used

3. EDCA (Enhanced Distributed Channel Access)

- Improved version of DCF with QoS support
- Traffic divided into categories:
 - Voice (highest priority)
 - Video
 - Best effort
 - Background
- Higher priority traffic gets faster access

9. Brief about the Block ACK mechanism and its advantages

Block ACK is used to acknowledge multiple frames at once instead of sending an ACK for each individual frame.

- Sender transmits a group of frames

- Receiver stores received frames
- Receiver sends a Block ACK frame
- Block ACK contains a bitmap indicating which frames were received successfully
- Sender retransmits only the missing frames

Advantages:

- Reduces number of ACK frames so less overhead
- Improves transmission efficiency
- Increases overall throughput
- Reduces delay in communication
- Works well with frame aggregation (A-MPDU)

10. Explain about A-MSDU, A-MPDU and A-MSDU in A-MPDU

These are frame aggregation techniques used to improve Wi-Fi performance by sending more data in one transmission.

A-MSDU (Aggregated MAC Service Data Unit):

- Combines multiple payloads into one frame
- Uses single MAC header
- Lower overhead, but if error occurs then whole frame retransmitted
More efficient, but less reliable

A-MPDU (Aggregated MAC Protocol Data Unit):

- Combines multiple full frames
- Each frame has its own MAC header and FCS
- Uses Block ACK for acknowledgement
More reliable, supports retransmission of only failed frames

A-MSDU in A-MPDU (Combined):

- Multiple A-MSDUs are packed inside A-MPDUs
- Combines benefits of both techniques
High efficiency + high reliability